AQ6.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Choose the set of correct options.

- [ ] If v_1 and v_2 are in the kernel of a linear transformation $T : V \to W$, where V and W are two vector spaces, then $v_1 + v_2$ will also be in the kernel of T.
- [ ] Kernel of a linear transformation $T : V \to W$ is a vector subspace of V, where V and W are two vector spaces.
- [ ] If w_1 and w_2 are in the image of a linear transformation $T : V \to W$, where V and W are two vector spaces, then $w_1 + w_2$ will also be in the image of T.
- [ ] Image of a linear transformation $T : V \to W$ is a vector subspace of W, where V and W are two vector spaces.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and v_2 are in the kernel of a linear transformation $T : V \to W$, where V and W are two vector spaces, then $v_1 + v_2$ will also be in the kernel of T.

Kernel of a linear transformation $T : V \to W$ is a vector subspace of V, where V and W are two vector spaces.

If w_1 and w_2 are in the image of a linear transformation $T : V \to W$, where V and W are two vector spaces, then $w_1 + w_2$ will also be in the image of T.

Image of a linear transformation $T : V \to W$ is a vector subspace of W, where V and W are two vector spaces.

Consider the following linear transformation:

$$T : R^3 \to R^2$$

$$T(x, y, z) = (2x + 3z, 4y + z)$$

Answer questions 2,3,4 and 5, using the information given above.

1 point

Which of the following matrices corresponds to the given linear transformation T with respect to the standard ordered basis for R^3 and the standard ordered basis for R^2 ?

- ( ) $\begin{bmatrix} 2 & 3 & 0 \\ 4 & 1 & 0 \end{bmatrix}$
- ( ) $\begin{bmatrix} 2 & 0 \\ 0 & 4 \\ 3 & 1 \end{bmatrix}$
- ( ) $\begin{bmatrix} 2 & 0 & 3 \\ 0 & 4 & 1 \end{bmatrix}$

○

$\begin{bmatrix}2&4\\3&1\\0&0\end{bmatrix}$ [230410]

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\begin{bmatrix}2&0&3\\0&4&1\end{bmatrix}$[200431]

1 point
Which of the following represents a basis of the kernel of TT?

○ $\{(-\frac{3}{2}, 0, 1), (0, -\frac{1}{4}, 1)\}$ {(−23, 0, 1), (0, −41, 1)}

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○ $\{(-\frac{3}{2}, -\frac{1}{4}, 2)\}$ {(−23, −41, 2)}

○ $\{(2, 0, 3), (0, 4, 1)\}$ {(2, 0, 3), (0, 4, 1)}

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(-\frac{3}{2}, -\frac{1}{4}, 1)\}$ {(−23, −41, 1)}

What will be the dimension of the subspace $Im(T)$Im(T)?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
1 point
Choose the correct option.

○ TT is an isomorphism.

○ TT is one to one but not onto.

○ TT is onto but not one to one.

○ TT is neither one to one, nor onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
TT is onto but not one to one.

1 point
Choose the set of correct options.

☐ There exists a linear transformation $T: R^2 \to R^2$T : R2 → R2 whose range is contained in the circle $x^2 + y^2 = 1$x2 + y2 = 1.

- [ ] There exists a linear transformation $T: R^2 \rightarrow R^2$ whose range equals the line $x = y$.
- [ ] There exists a linear transformation $T: R^2 \rightarrow R^2$ whose range is contained in the set $S = \{(x, y) | x > 0\}$.
- [ ] There exists a linear transformation $T: R^2 \rightarrow R^2$ whose range equals the line $y = \pi x$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
There exists a linear transformation $T: R^2 \rightarrow R^2$ whose range equals the line $x = y$.
There exists a linear transformation $T: R^2 \rightarrow R^2$ whose range equals the line $y = \pi x$.

Level 2
1 point
Which option represents the kernel and image of the following linear transformation?

$T: R^2 \rightarrow R^2$
$T(x, y) = (x, 0)$

○ $ker(T) = Span\{(1, 0)\}, Im(T) = Span\{(1, 0)\}$
○ $ker(T) = Span\{(1, 0)\}, Im(T) = Span\{(0, 1)\}$
○ $ker(T) = Span\{(0, 1)\}, Im(T) = Span\{(1, 0)\}$
○ $ker(T) = Span\{(0, 1)\}, Im(T) = Span\{(0, 1)\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$ker(T) = Span\{(0, 1)\}, Im(T) = Span\{(1, 0)\}$

1 point
Which of the following linear transformations is an isomorphism between the vector spaces $V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in R\} \subset R^3$ and $W = \{(x, y, z) \mid x = y, \text{ and } x, y, z \in R\} \subset R^3$?

○ $T: V \rightarrow W$, such that $T(x, y, z) = (y, y, 0)$
○ $T: V \rightarrow W$, such that $T(x, y, z) = (y, y, z)$
○ $T: V \rightarrow W$, such that $T(x, y, z) = (x, y, z)$
○ There does not exist any isomorphism between V and W.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$T: V \to W$, such that $T(x, y, z) = (y, y, z)$

1 point

Which of the following linear transformations is an isomorphism between the vector spaces $V = \{(x, y, z) \mid x = y - z = 0, \text{ and } x, y, z \in R\} \subset R^3$ and $W = \{(x, y, z) \mid x = y = 0, \text{ and } x, y, z \in R\} \subset R^3$?

○ $T: V \to W$, such that $T(x, y, z) = (x, 0, y)$.

○ $T: V \to W$, such that $T(x, y, z) = (0, y, z)$.

○ $T: V \to W$, such that $T(x, y, z) = (0, 0, x)$.

○ There does not exist any isomorphism between V and W.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T: V \to W$, such that $T(x, y, z) = (x, 0, y)$.

1 point

Consider the linear transformation $T: R^3 \to R^3$ such that $T(x, y, z) = (x - y + z, 2x + y - z, -x - 2y + 2z)$. Which of the following options are true about T?

☐ Range of T is $\{(x, y, z) \in R^3 \mid x = y + z\}$.

☐ Range of T is $\{(x, y, z) \in R^3 \mid x = 0, y + z = 0\}$.

☐ Rank of T is 2

☐ Rank of T is 1

No, the answer is incorrect.

Score: 0

Accepted Answers:

Range of T is $\{(x, y, z) \in R^3 \mid x = y + z\}$.

Rank of T is 2

Check Answers

Your score is: 0/10